Clinical Risk Assessment Report: Patient 75

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 8.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.40x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.2%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.25 [Avg (36th %ile)] (-2.2%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.55 [Top 6%] (-1.1%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.5 [Avg (34th %ile)] (-1.0%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Patient Age (Age) **Value: 75.0 [Top 11%] (-1.0%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.41 [Bottom <1%] (-0.9%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.25 [Top 5%] (-0.7%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET) **Value: -0.45 [Bottom <1%] (-0.7%)**

Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Carcinoembryonic Antigen (CEA) **Value: 1.5 [Bottom 15%] (-0.6%)**

Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Machine Learning Feature Attribution (SHAP Decision Path)